

CSC324 Principles of Programming Languages

University of Toronto

Fall 2026

About this course

The study of programming languages is unique in that it blends the practical skill of translating a problem into a solution that a computer can run, with theoretical underpinnings and formalism from the mathematics community.

By the end of the course you should be able to:

- read and write programs in the λ -calculus, and reason about how they reduce;
- write functional programs in Haskell using datatypes, recursion, higher-order functions and type classes;
- explain how a language is implemented by building an interpreter for a small language;
- say which features of a language are *fundamental* and which are syntactic sugar, and use that to pick up an unfamiliar language quickly.

Prerequisites

CSC263 or CSC265 is required for this course. Specifically, this translates to you needing to have:

- Familiarity with common data structures such as linked lists and trees;
- Competency in some prior programming language, such that you could implement one of the above data structures;
- Comfort with recursion, and understanding and writing proofs by induction.

In essence, if you're reasonably convinced that you could design a binary tree datatype and write a recursive function/method that traverses it, and explain in your own words why your solution is correct, you will be in good shape for this class.

Teaching team

Instructor: Prof. Ningning Xie, BA 3256 — office hour Monday 11:30–12:30

Teaching assistants: Andong Fan, Matthew Toohey, Yanning Chen, Oghenevwogaga Ebresafe — office hour Friday 2–3pm, BA 2230

The TAs run tutorials, hold weekly office hours, and answer questions on Piazza. Office hours begin in week 2.

Meeting times

Lectures: Tuesday 9–11am and 3–5pm

Tutorials: Thursday 10–11am and 3–4pm

Course calendar: subscribe to the course calendar for all lecture, tutorial, deadline and exam dates. The calendar is an authoritative source for dates.

Where things live

- **Quercus** (q.utoronto.ca) — course website, lecture slides, tutorial materials, and official announcements. *It is your own responsibility to check it at regular intervals, i.e. once per day.* It is strongly recommended that you select “Notify immediately” for course announcements in your account notification settings, which forwards all announcements to your university email.
- **MarkUs** — all assignments are submitted here.
- **Piazza** — questions about lecture and assignment *content*. Please do not email instructors or TAs about course material; ask on Piazza, or in lectures, tutorials and office hours. Staffed Monday–Friday during working hours.
- **Course email** (csc324-2026-09@cs.toronto.edu) — personal and logistical questions only. We run a ticket system on this address; please avoid emailing the instructor or TAs directly.

No textbook. We follow the slides; useful resources will be posted on Quercus.

Topics

- The lambda calculus (∼2–3 weeks)
- Functional programming and types (∼5 weeks)
- Building an interpreter (∼2 weeks)
- Overloading and typeclasses (∼2 weeks)

Grading scheme

	Weight	Due
Assignment 1	10%	Sept 30
Assignment 2	10%	Nov 4
Assignment 3	10%	Dec 2
Midterm	25%	Oct 13
Final	45%	tbd
Piazza participation	5% (extra credit)	—

The 5% Piazza extra credit goes to the top five participants over the term.

Autofail: you must earn at least **35%** of the final to pass the course.

Assignments

No groups. Discussion is fine; the solution you submit must be your own.

Flex tokens. Every student receives ten flexible tokens, each of which extends a deadline by 12 hours. Multiple tokens may be used on one assignment. Tokens are applied automatically on MarkUs — no request is needed.

Additional accommodations. The token system should be enough for most situations. If you need more, email the course email (csc324-2026-09@cs.toronto.edu) as soon as possible and at least **12 hours** before the deadline, not counting flex tokens, with documentation:

- Absence Declaration via ACORN
- U of T Verification of Illness or Injury Form (VOI)
- College Registrar’s letter
- Letter of Academic Accommodation from Accessibility Services

Maximum 3 days grace per assignment, with flex tokens and accommodations combined. Late submissions without properly used flex tokens or an approved extension will not be graded and automatically receive 0 marks.

We release assignment solutions once everyone has submitted, or after the due date + 3 days.

Regrading

- We do **not** accept regrading requests in the first **24 hours** after grades are released. You should spend this time reading and understanding the solutions and your assessment results.
- After that you have **3 days** to submit a regrading request, with justification for why regrading is needed.

Exams

- Closed book; the exam itself will include a cheatsheet.
- The final is cumulative.
- You must earn at least 35% of the final to pass the course.

Tutorials

Tutorials recap the lecture content and work through exercises.

- Exercises are released Tuesday at 6pm, for you to practise on.
- TAs talk through the exercises in Thursday's tutorial.
- Solutions are released Thursday at 6pm.

Academic integrity

Cheating is against “fair play” and will not be tolerated under any circumstances. While the pressures of many classes, homework, work and/or extracurricular activities can be great, this is never an excuse for copying solutions from others. **The University holds among its highest principles the notion of academic freedom and integrity. Cheaters will face the University's disciplinary committee as well as receive a failing grade in this course.** If you think there is an issue affecting your performance in the class, talk to the instructor.

Concretely, for this course:

- Use Piazza for questions and answers.
- Do **not** use generative AI technology for assignments.
- Do **not** post full or partial solutions to assignments on Piazza.
- Do **not** share your test cases — learning to test is an important part of programming.
- Please don't “help” others by helping them do their work, and please don't put others in an awkward position by asking them for help on an assignment. If someone asks you for help, ask them to post a question on Piazza for you to answer.

Accessibility

If you have a disability or health consideration that may require accommodations, please approach Accessibility Services as soon as possible. Letters of Academic Accommodation are honoured in this course; see the assignment section above for how they interact with deadlines.